A RING COMPLETE FOR A PRINCIPAL IDEAL WITH A PRINCIPAL IDEAL WHOSE QUOTIENT IS NOT COMPLETE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds no mathematics of its own; the complete chain (canonical form, replacement rules, contradiction) lives in the context span.

Editorial note. The label lemma-noncomplete-quotient names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 680-727: R = k[t, z_1, z_2, ..., w_1, w_2, ..., x]/(z_i t - x^i w_i, z_i w_j); every element has a unique canonical form f = sum_{i=0..d} f_i x^i with f_i in k[t, z_i, w_j] avoiding the products z_i t and z_i w_j; f in (x^n) iff f_i = 0 for i < n; so x is a nonzerodivisor and the intersection of the (x^n) is 0
- The (x)-adic completion R^wedge consists of unique infinite sums f = sum_{i>=0} f_i x^i (Algebra, hathat-finitely-generated); the element f = sum_{i>=1} x^i w_i lies in (t, x^n) for every n because x^m w_m is in (t); the claim f not in (t) is proven by the replacement rules for multiplying t g in canonical form (a term of g_0 involving z_i contributes c w_i to f_{l+i}; picking n with z_n absent from the polynomial g_0 leaves w_n absent from f_n, a contradiction)
- Algebra, quotient-complete: a quotient of a complete ring is complete iff it is classically complete, so R^wedge/(t) is not complete

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Prerequisite pointer invoked at 755; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031A: sources/stacks-chow/tags.txt, source lines 3977–3977. Prerequisite pointer invoked at 786; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let k be a field. Let

$$R = k[t, z_1, z_2, z_3, \ldots, w_1, w_2, w_3, \ldots, x]/(z_i t - x^i w_i, z_i w_j)$$

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Note that in particular $z_i z_j t = 0$ in this ring. Any element f of R can be uniquely written as a finite sum

$$f = \sum\nolimits_{i=0,\ldots,d} f_i x^i$$

where each $f_i \in k[t, z_i, w_j]$ has no terms involving the products $z_i t$ or $z_i w_j$. Moreover, if f is written in this way, then $f \in (x^n)$ if and only if $f_i = 0$ for $i < n$. So x is a nonzerodivisor and $\bigcap(x^n) = 0$. Let $R^\wedge$ be the completion of R with respect to the ideal (x). Note that $R^\wedge$ is (x)-adically complete, see Algebra, Lemma 05GG. By the above we see that an element of $R^\wedge$ can be uniquely written as an infinite sum

$$f = \sum\nolimits_{i=0}^{\infty} f_i x^i$$

where each $f_i \in k[t, z_i, w_j]$ has no terms involving the products $z_i t$ or $z_i w_j$. Consider the element

$$f = \sum\nolimits_{i=1}^{\infty} x^i w_i = x w_1 + x^2 w_2 + x^3 w_3 + \ldots$$

i.e., we have $f_n = w_n$. Note that $f \in (t, x^n)$ for every n because $x^m w_m \in (t)$ for all m. We claim that $f \notin (t)$. To prove this assume that $tg = f$ where $g = \sum g_l x^l$ in canonical form as above. Since $t z_i z_j = 0$ we may as well assume that none of the g_l have terms involving the products $z_i z_j$. Examining the process to get tg in canonical form we see the following: Given any term cm of g_l where $c \in k$ and m is a monomial in t, z_i, w_j and we make the following replacement

(1) if the monomial m does not involve any z_i, then ctm is a term of f_l, and
(2) if the monomial m does involve a z_i then it is equal to $m = z_i$ and we see that cw_i is term of f_{l+i}.

Since g_0 is a polynomial only finitely many of the variables z_i occur in it. Pick n such that z_n does not occur in g_0. Then the rules above show that w_n does not occur in f_n which is a contradiction. It follows that $R^\wedge/(t)$ is not complete, see Algebra, Lemma 031A.

Lemma 0.1. *There exists a ring R complete with respect to a principal ideal I and a principal ideal J such that R/J is not I-adically complete.*

Proof. See discussion above. □